

AMENDMENT NO. 1 AUGUST 2024
TO
IS 17292 : 2020 PUMPS — MINIMUM REQUIRED EFFICIENCY OF
ROTODYNAMIC WATER PUMPS

[Page 4, clause 6.1, b), 1), last line] — Substitute ‘Annex C’ for ‘Annex B’.

(Page 5, clause 6.2, para 3, line 3) — Substitute ‘Annex D’ for ‘Annex C’.

(Page 9, clause 7.3, para 1) — Insert the following at the end of para.

‘,especially run-in effects shall be considered.

NOTE — Run-in effects may take up to one-day operating time.’

(Page 19, Annex B) — Substitute the following for the existing:

‘ANNEX B

(Clause 2)

LIST OF REFERRED STANDARDS

<i>IS No./ Other Standards</i>	<i>Title</i>
IS 9137 : 2019	Code for hydraulic performance acceptance tests for centrifugal, mixed and axial flow pumps — Class C (<i>first revision</i>)
ISO 9906 : 2012	Rotodynamic pumps — Hydraulic performance acceptance tests — Grades 1, 2 and 3
IS/ISO 80000 (Part 1) : 2022	Quantities and units: Part 1 General’.

[Page 19, Annex D, para 7 b)] — Substitute ‘percent to’ for ‘percentto’.

(Page 28) — Insert the following new clause after **F-1.7**.

‘F-1.8 It can clearly be seen from the example that the width of the 95 percent confidence intervals of the mean efficiency of the same size and determined with the same measurement accuracy are much larger when testing only one sample pump than when testing a sample of five pumps (it has to be noted that in the example the default value of $t_{\text{man},\eta} = \pm 0.04$ describes exactly the actual scatter of efficiencies of individual pumps within the size).

Finally, the results shall serve to prove that for the size a Minimum Efficiency Index of $\text{MEI} = 0.4$ can be indicated. According to **6**, the value of η_{mean} has to be compared to the corresponding minimum required value. To fulfill the qualification criterion for $\text{MEI} = 0.4$, the value of η_{mean} shall be equal to or higher than the corresponding minimum required value. To determine the minimum required values of η_{mean} , first the nominal values of minimum required efficiency are calculated by the means of Formula (4) for $\text{MEI} = 0.4$ with the values of:

- a) $Q_{\text{BEP,test}}$ and $\eta_{\text{s,test}}$ in the case of $M = 1$; and
- b) $Q_{\text{BEP,mean}}$ and $\eta_{\text{s,mean}}$ in the case of $M = 5$ as input quantities.

Price Group 2

AMD 1 to IS 17292 : 2020

According to **8**, the value of η_{mean} within its 95 percent confidence interval has to be compared to the corresponding threshold value. To fulfill the verification criterion for $\text{MEI} = 0.4$, the value of η_{mean} shall be equal to or higher than the corresponding particular threshold value. To determine the particular threshold values of η_{mean} , the previous values resulting from Formula (4), is multiplied by the factor 0.95 (which corresponds to the total tolerance $t\eta = -5$ percent as per ISO 9906, Class 2). The mentioned values are shown in Table F-5. In the last column of Table F-5, also the result of the comparison in respect to the fulfillment of the qualification and verification criterion for $\text{MEI} = 0.4$.

Table F-5 Relevant Values and Results in Respect to Qualification of the Size

(Clause F-1.8)

Sl No.	No. of Test Pump	Measured Efficiency	Lowest Value of 95 Percent Confidence Interval	Minimum Required Value	Particular Threshold Value	Fulfilled Qualification Criterion	Fulfilled Verification Criterion
	i	η_{EIP}	$\eta_{\text{EIP}} - t_{\text{tot}}$	$\eta_{\text{EIP,requ.}}$	$\eta_{\text{EIP,threshold}}$		
		(Percent)	(Percent)	(Percent)	(Percent)		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	1	58.70	55.78	60.90	57.86	No	Yes
ii)	2	61.13	56.89	60.77	57.73	Yes	Yes
iii)	3	61.92	58.70	60.22	57.20	Yes	Yes
iv)	4	59.90	56.87	60.47	57.45	No	Yes
v)	5	62.05	58.38	60.83	57.78	Yes	Yes
vi)	All test pumps combined	60.74	58.66	60.64	57.61	Yes	Yes

Although in this example, the true mean value $\eta_{\text{BEP,mean}} = 60.74$ percent is actually higher than the nominal minimum required values (qualification criterion), the results for two of the five pumps – when taken as a single test sample – would not prove that a value of $\text{MEI} = 0.4$ is justified. On the other hand, when all five pumps are taken as a sample of $M = 5$, the result (qualification criterion) would not only prove clearly that a value of $\text{MEI} = 0.4$ is justified.

NOTE — The indication of the value $\text{MEI} = 0.4$ for the size, proven by the manufacturer on the basis of the sample of $M = 5$ pumps, would be confirmed by an independent institution (applying the procedure described in **8**) already with the first test if anyone of the five pumps serving for the example above would be drawn at random out of the size.

Only in cases of very low probability if only one pump is tested for the qualification, a pump chosen for the verification would fail. But also in this cases, the indicated value of $\text{MEI} = 0.4$ would be confirmed with very high probability by tests on three additional sample pumps (see Annex G).’

(Page 31, Table G-2) — Substitute the following for the existing:

Table G-2 Maximum Permissible Values of Ratio R (Two Sided 95 Percent)

(Clause G-2.1)

Sl No.	N or M	R_{max}
(1)	(2)	(3)
i)	3	1.15
ii)	4	1.48
iii)	5	1.71
iv)	6	1.89

SI No.	<i>N</i> or <i>M</i>	<i>R</i> _{max}
(1)	(2)	(3)
v)	7	2.02
vi)	8	2.13
vii)	9	2.21
viii)	10	2.29
ix)	11	2.36
x)	12	2.41
xi)	13	2.46
xii)	14	2.51
xiii)	15	2.55
xiv)	16	2.59
xv)	17	2.62
xvi)	18	2.65
xvii)	19	2.68
xviii)	20	2.71
xix)	21	2.73
xx)	22	2.76
xxi)	23	2.78
xxii)	24	2.80
xxiii)	25	2.82
xxiv)	26	2.84
xxv)	27	2.86
xxvi)	28	2.88
xxvii)	29	2.90
xxviii)	30	2.91'

(Page 32, clause **G-3**, para 2, equation G-8) — Substitute the following for the existing:

$$S_y = \sqrt{\sum_{i=1}^n \left(\frac{dy}{dx_i}\right)^2 \times S_{x_i}^2} \dots\dots\dots(\text{G-8})'.$$

(Page 32, clause **G-3**, para 3, equation G-9) — Substitute the following for the existing:

$$S_y = \sqrt{\sum_{i=1}^n S_{x_i}^2} \dots\dots\dots(\text{G-9})'.$$

[Page 33, clause **G-4.5**, d)] — Substitute the following for the existing:

‘d) Same values of systematic uncertainties (that is, same accuracy of test equipment) are assumed for all test pumps of the sample:

- 1) *e*_{s,Q}: 0.01 p.u. =1 percent;
- 2) *e*_{s,H}: 0.01 p.u. =1 percent;
- 3) *e*_{s,n}: 0.01 p.u. =1 percent; and
- 4) *e*_{s,T}: 0.01 p.u. =1 percent.’

[Page 33, Table G-5, col (7), SI No. ii)] — Substitute ‘57.67 – 64.59’ for ‘57.67 – 60.59’.

[Page 36, clause **H-1**, b), equation H-3] — Substitute the following for the existing:

$$e_{r,x} = \bar{x} \pm 1.96. s_x / \sqrt{N} \text{ for } N \geq 30 \dots\dots\dots(\text{H-3})'.$$

AMD 1 to IS 17292 : 2020

[Page 36, clause **H-1**, b), equation H-4] — Substitute the following for the existing:

$$e_{r,x} = \bar{x} \pm t(P, N) \cdot s_x / \sqrt{N} \text{ for } N < 30 \dots\dots\dots (\text{H-4}).$$

[Page 37, clause **H-2**, a), equation H-9] — Substitute the following for the existing:

$$e_{tot,\eta} = \sqrt{e_{tot,Q}^2 + e_{tot,H}^2 + e_{tot,n}^2 + e_{tot,T}^2 \dots\dots\dots} (\text{H-9}).$$

[Page 37, clause **H-2**, b), equation H-10] — Substitute the following for the existing:

$$e_{tot,\eta} = \sqrt{e_{tot,Q}^2 + e_{tot,H}^2 + e_{tot,Pgr}^2 + e_{tot,\eta dr}^2 \dots\dots\dots} (\text{H-10}).$$

(Page 38, clause **I-1**, para 3, line 6) — Substitute ' $\pm t_{tot,\bar{\eta}} \cdot \bar{\eta}$ ' for ' $\pm t_{tot,\eta} \cdot \bar{\eta}$ '.

(Page 38, clause **I-1**, para 7, line 4) — Substitute ' $\bar{\eta} - \eta_{threshold} = + |t_{tot,\eta}| \cdot \bar{\eta}$ ' for ' $\bar{\eta} - \eta_{threshold} = + |t_{tot,\eta}| \cdot \eta$ '.

(Page 39, clause **I-1**, para 8) — Substitute ' $\bar{\eta} - \eta_{threshold} = - |t_{tot,\eta}| \cdot \bar{\eta}$ ' for ' $\bar{\eta} - \eta_{benchmark} = - |t_{tot,\eta}| \cdot \eta$ '.

(Page 41, Table **J-1**, row 22) — Substitute ' η ' for ' h '.

(Page 41, Table **J-1**, row 25, 26) — Substitute ' l/s ' for ' m^3/hr '.

(Page 41, Table **J-1**, row 29) — Substitute ' η_t ' for ' h_t '.

(Page 41, Table **J-1**, row 31) — Substitute ' C_{right} and C_{left} ' for ' C '.

(MED 20)

Publication, BIS, New Delhi